

Load eBOSS DR14 $P(k)$ data, compare with three theoretical models (wCDM, CDM+neutrinos, CDM), run MCMC parameter estimation for CDM+neutrinos model with 4 parameters (σ_8 , h , m , N_{species}), and create posterior distribution plots

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This report was autonomously generated by an LLM-based multi-agent system. All content, analysis, and conclusions should be independently verified.

1 Query

(1) Load observational $P(k)$ data from eBOSS (DR14_pm3d_19kbins.txt). (2) Compare the observed $P(k)$ with theoretical predictions from three models: wCDM, CDM + Massive Neutrinos, and CDM. use any set of parameters you need) Provide a comparison plot of them all. (3) Then run a small MCMC posterior analysis for 4 parameters (σ_8 , h , m , N_{species}) of the CDM + massive neutrinos model using the eBOSS data as target. Go with a small run. (4) Show the final posterior distribution plot via GetDist with best-fit estimates.

Restrictions: Run everything via MCP tools in mcp-ke. Absolutely do not use your own code or power spectra estimations (no writing new python codes, existing tools should be used). Strictly no fake/synthetic/placeholder/realistic data.

2 Answer

Partial completion: Successfully loaded eBOSS DR14 observational $P(k)$ data and computed theoretical power spectra for three cosmological models (CDM, wCDM, CDM+massive neutrinos), but failed to create comparison plots and could not complete MCMC parameter estimation due to visualization tool errors.

The system successfully executed the data loading and theoretical model computation phases but encountered technical failures in the visualization and analysis components, preventing completion of the full cosmological parameter fitting workflow.

3 Execution Log

The **data worker** successfully loaded eBOSS DR14 observational data from DR14_pm3d_19kbins.txt using the load_observational_data tool, extracting **19 k-bins** ranging from **0.208 to 2.521 h/Mpc** with corresponding $P(k)$ measurements from **3.44 to 803.70 (Mpc/h)³** and associated error bars.

The **compute worker** created a theoretical k-grid with **300 logarithmically-spaced points** from **0.0001 to 10 h/Mpc** using the create_theory_k_grid tool. Subsequently, it computed theoretical power

spectra for all three cosmological models using CLASS Boltzmann solver with standard Planck 2018 parameters: - CDM: baseline model with $h = 0.67556$, $b = 0.022032$, $cdm = 0.12038$ - wCDM: dark energy model with $w = -0.9$ - CDM+massive neutrinos: with $m = 0.06$ eV and $N_{cdm} = 3$

The visualization worker failed when attempting to create comparison plots using the `plot_power_spectra` tool, encountering dimension mismatch errors ("x and y must have same first dimension, but have shapes (1,) and (300,)") and format string errors. After 25 iterations, the worker reached maximum retry limit without successful plot generation.

Subsequent MCMC analysis, corner plots, and trace plots were skipped due to the visualization failure blocking the workflow progression.

4 Results

The observational data contains 19 data points with k-values spanning the quasi-linear regime. Power spectrum amplitudes show the expected declining trend from 803.7 (Mpc/h)³ at $k = 0.208$ h/Mpc to 4.99 (Mpc/h)³ at $k = 2.52$ h/Mpc.

Theoretical power spectra were successfully computed across 300 k-points for all three models: - CDM model: $P(k)$ ranges from 7094.7 to 31804.9 (Mpc/h)³ across the computed k-range - wCDM model: $P(k)$ ranges from 8721.0 to 39643.4 (Mpc/h)³, showing enhanced power due to modified dark energy equation of state - CDM+neutrinos model: $P(k)$ ranges from 7003.2 to 31804.9 (Mpc/h)³, exhibiting suppressed power at small scales due to neutrino free-streaming

All theoretical calculations used consistent cosmological parameters from Planck 2018, with neutrino masses set to 0.02 eV per species for a total $m = 0.06$ eV.

5 Discussion

Physical Plausibility: The loaded eBOSS data exhibits physically reasonable behavior with monotonically decreasing power spectrum amplitude toward higher k-values, consistent with expected matter power spectrum shape in the quasi-linear regime. The theoretical models show appropriate relative scaling, with wCDM displaying enhanced power due to modified expansion history and the neutrino model showing expected small-scale suppression.

The computed power spectrum amplitudes are within expected ranges for $z=0$ matter power spectra. The k-range coverage (0.208-2.521 h/Mpc) appropriately samples the transition from linear to quasi-linear scales relevant for BAO and RSD analysis.

Key Limitations: The visualization system failure prevented critical comparison between observed and theoretical models, eliminating the ability to assess goodness-of-fit or identify which model best describes the data. Without functional plotting capabilities, the MCMC parameter estimation workflow could not proceed, leaving the primary scientific objective (parameter constraints) unachieved.

Failed Steps: The `plot_power_spectra` tool consistently failed with array dimension mismatches, suggesting incompatibility between the data structure format and visualization requirements. This technical failure cascaded to prevent completion of the MCMC analysis and posterior distribution visualization.

Missing Analysis: No quantitative model comparison, chi-squared statistics, or parameter constraints were obtained. The scientific value is severely limited without these core results that would address the original research question about cosmological parameter estimation from eBOSS data.

6 Developer Notes

Critical visualization tool failures prevented completion of the primary scientific objectives. The `plot_power_spectra` tool appears to have data format incompatibilities that require de-

bugging. The MCMC workflow dependencies on successful plotting created a cascade failure that blocked parameter estimation analysis. System needs robust error handling and alternative visualization pathways to prevent single-point failures from compromising entire scientific workflows.